



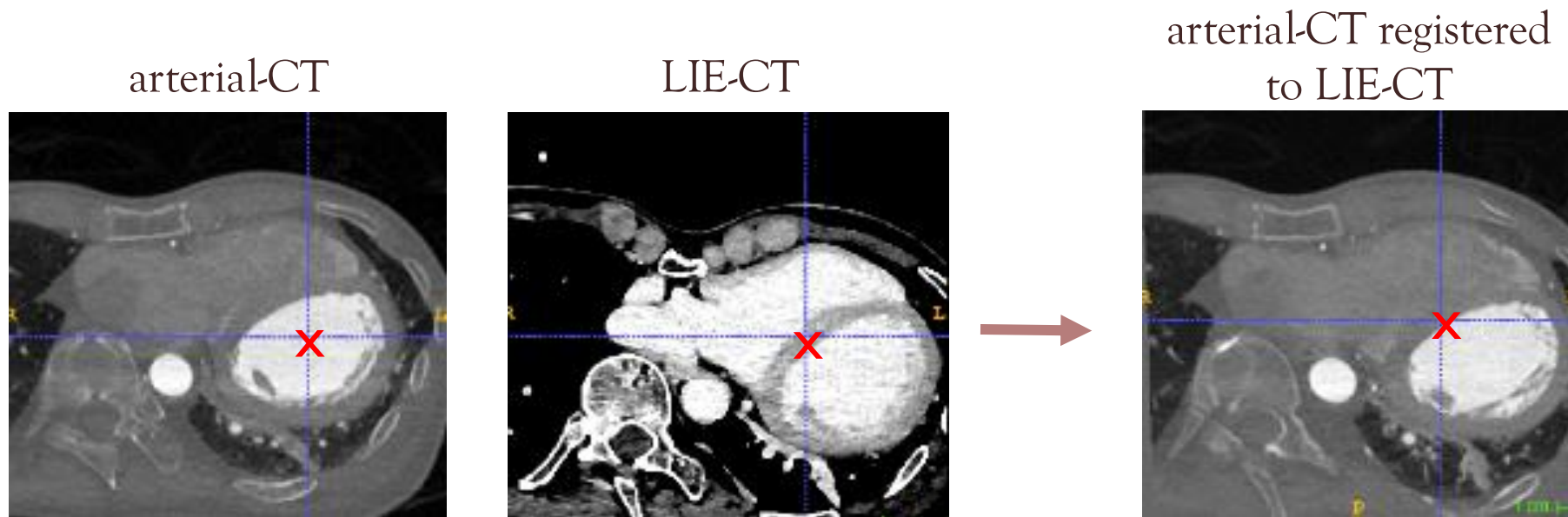
6. MATCHING

1. *Corner detector*
2. *Blob detector*
3. *Matching algorithms*

MATCHING FEATURES: WHY?

Establish **correspondence** between 2 images for:

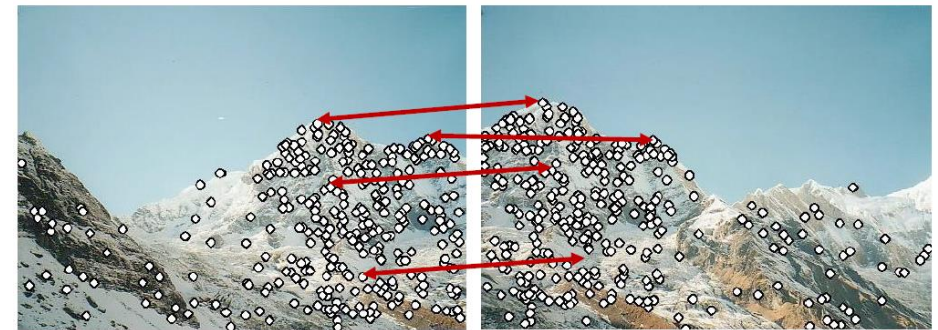
- **Image registration** based on feature correspondence: **transform** image to minimize similarity metric



MATCHING FEATURES: WHY?

Establish **correspondence** between 2 images for:

- **Image registration** based on feature correspondence
- **Panorama Stitching:** 2 images, how do we combine them?
 1. **Extract** keypoints
 2. **Match** keypoint features
 3. **Align** images



MATCHING FEATURES

Useful in many more CV **applications:**

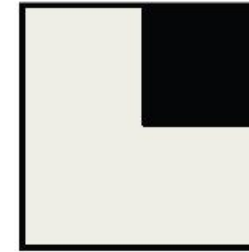
- Image alignment
- Object detection
- Object recognition
- Motion tracking
- 3D object reconstruction
- Structure-from-motion
- Image retrieval
- Point matching for computing disparity
- Stereo calibration (estimation of the fundamental matrix)
- Motion-based segmentation
- Robot navigation
- etc.

MATCHING: KEYPOINTS

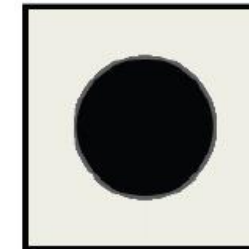
- From edge detection (Topic 3) to **keypoints**: features in specific locations (e.g. corners)
- Often described by the **appearance of patches of pixels** surrounding the point location



Corners



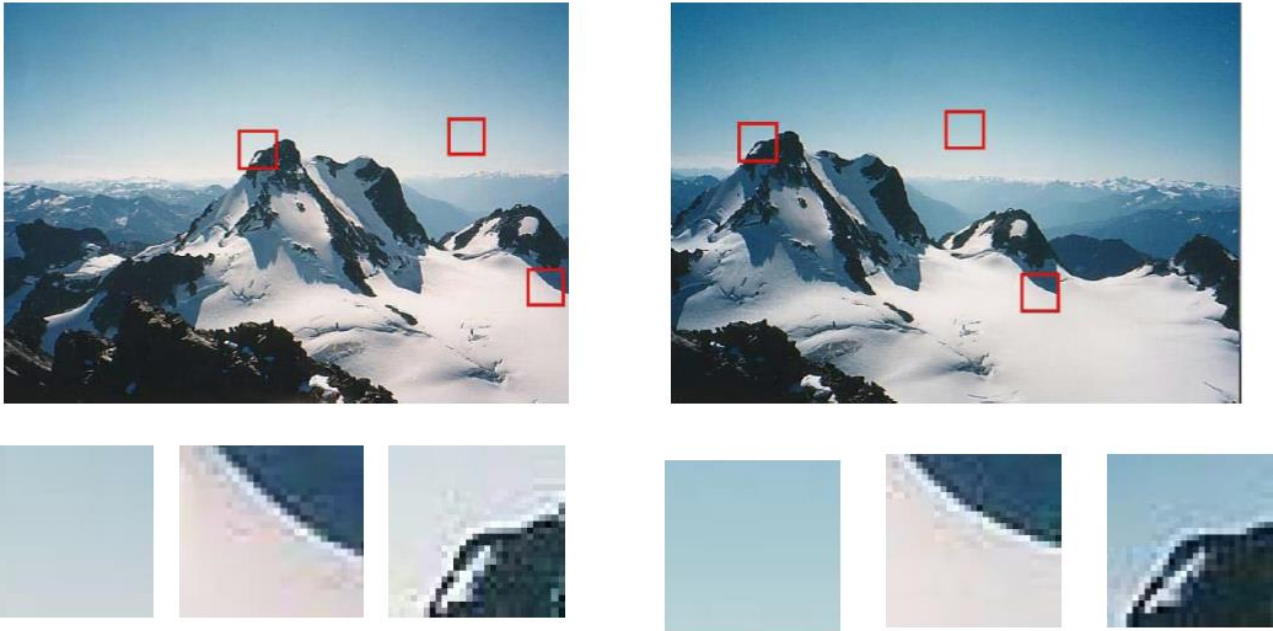
Blobs



MATCHING FEATURES

Best features for matching? → The most **distinctive**

Example: 2 images, same thing, 2 different **perspectives**



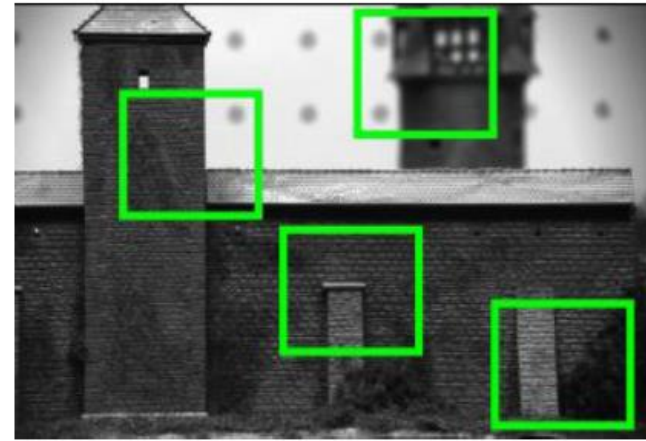
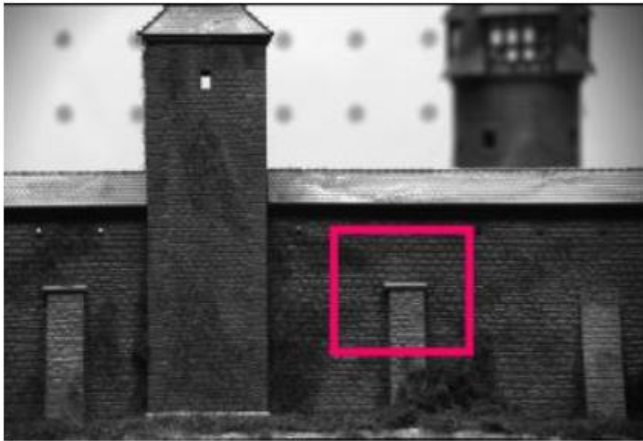
Szeliski, 2010

Characteristics of good keypoints:

1. **Repeatability:** Can be found despite geometric and photometric transformations
2. **Salience:** Each keypoint is distinctive
3. **Compactness and efficiency:** Many fewer points than image pixels
4. **Locality:** Occupies small area of the image; but robust to clutter and occlusion

MATCHING FEATURES

Best features for matching? → The most **distinctive**



Collings

Only one
looks similar

MATCHING FEATURES

Best features for matching? → The most **distinctive**

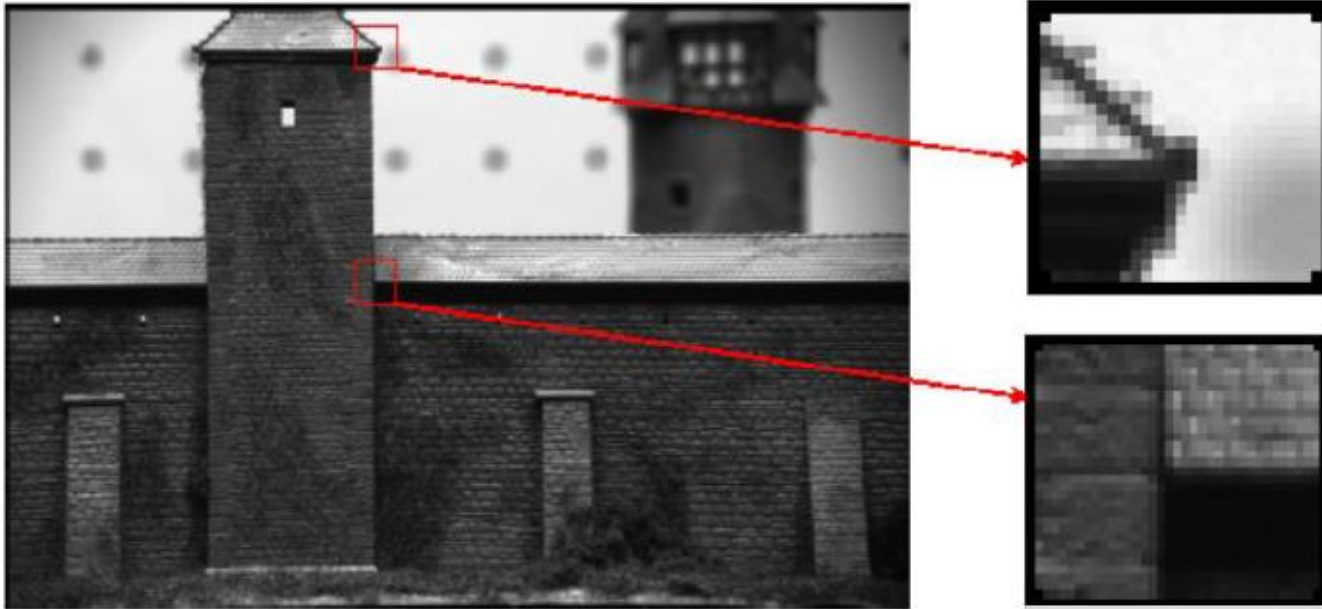


Collings

Many similar patches

MATCHING FEATURES

Best features for matching? → **Corners?**



Collings

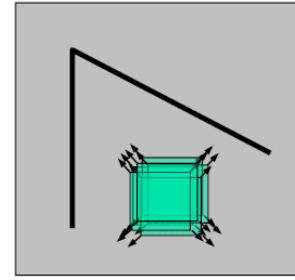
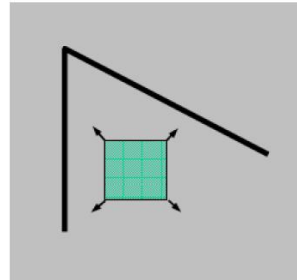
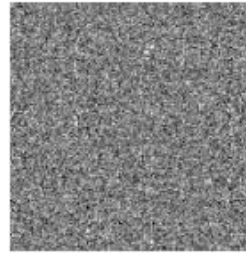
Corner = large variation in its neighborhood

Good features to match because:

- **Distinctive**
- **Reasonably invariant** to viewpoint
- **Easy to locate** → contour junctions

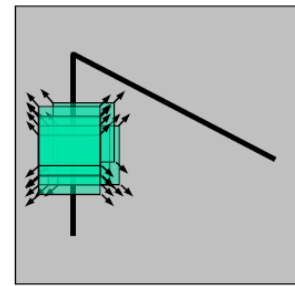
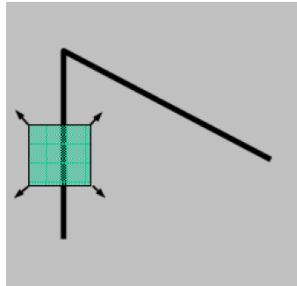
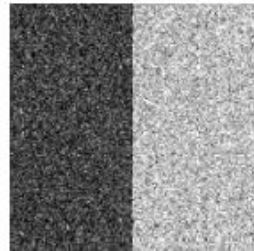
MATCHING FEATURES

Flat regions



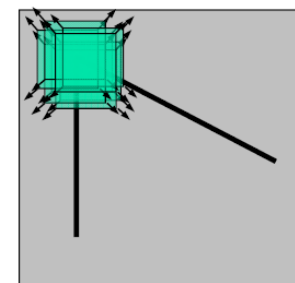
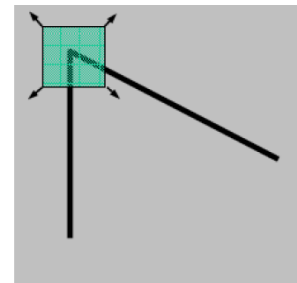
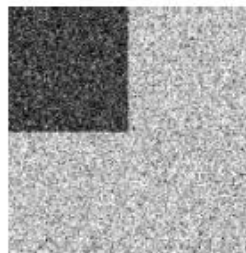
No change in all directions

Edges



No change along the edge direction

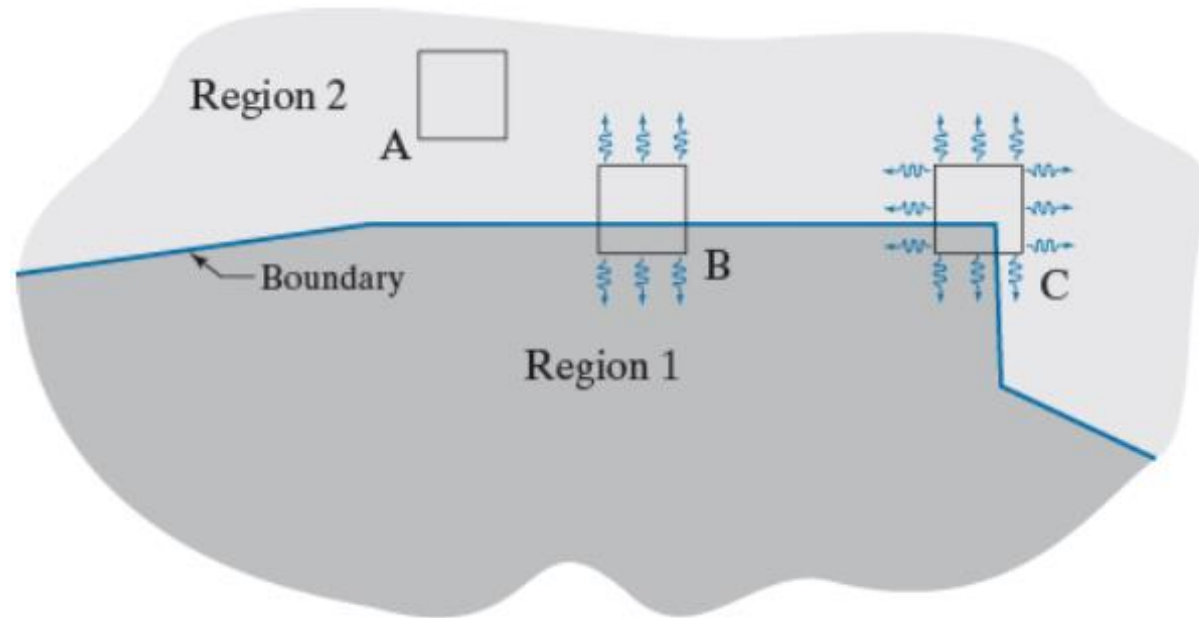
Corners



Significant changes in all directions

MORAVEC DETECTOR

- One of earliest corner detection algorithm (1980)
- Defines a corner to be a **point with low self-similarity**
- Corners are detected by **running a small window** over an image & computing **intensity changes**
- 3 scenarios:
 - Region A (flat)
 - Region B (edge)
 - Region C (corner)



Harris, C. and Stephens, M. [1988]. "A combined corner and edge detection", *Proc. 4th Alvey Vision Conf*, pp. 240-246.

MORAVEC DETECTOR: MATHS

Weighted sum of squared differences between the 2 patches:

$$E_{m,n}(x, y) = \sum_{u,v} \underbrace{w_{m,n}(u, v)}_{\text{Window function}} \underbrace{[f(u+x, v+y) - f(u, v)]^2}_{\substack{\text{Shifted intensity} \\ \text{Intensity}}} \quad \forall m, n, x, y$$

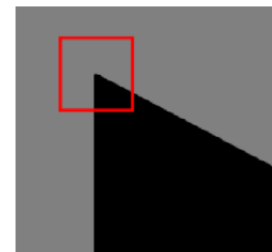
f : image

$f(u, v)$: patch of the image defined by the area (u, v)

$f(u+x, v+y)$: same but shifted by (x, y)

(x, y) represents the **offsets/shift** in the 4 directions
 $w_{m,n}$: rectangular window around the (m, n) pixel

- Nearly constant patches $\rightarrow$ close to 0
- Very distinctive patches $\rightarrow$ large value



- Lower values $\rightarrow$ more similarity
- For each pixel, $F_{m,n} = \min(E_{m,n}(x, y))$, the minimum in the 4 directions
- **Corners** correspond to local **maxima** of $F_{m,n} \rightarrow$ pixels where the smallest value of $E_{m,n}(x, y)$ is **large**

MORAVEC DETECTOR

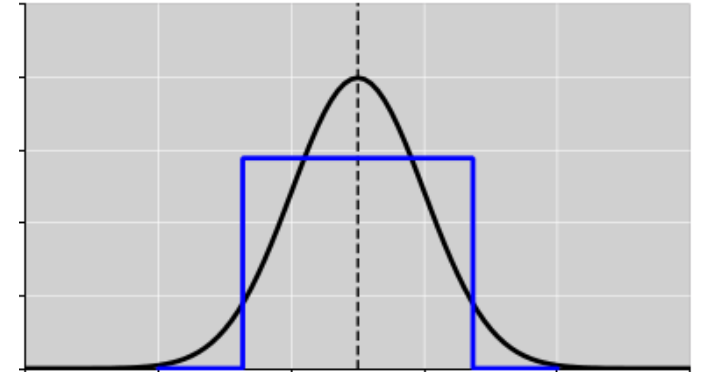
Drawbacks:

- $w_{m,n}$, rectangular, **binary window**:
 - The detector considers all pixels in the window with the **same weight**
 - It may be interesting that the closest ones have higher weight
 - If noisy images $\rightarrow$ false positive corner detections
- Only 4 directions are considered $\rightarrow$ **no rotation invariance**
- Sensitive to **edges** (false positives) because only the minimum of E (only 1 direction in the end) is considered

HARRIS-STEPHENS DETECTOR

Improved version of the Moravec detector:

1. Replace **rectangular window**, w , by a **Gaussian window** → noisy response reduction
2. Extension to **all directions**, not limited to the initial 4 directions. A Taylor series expansion is performed on the shifted sub-image $f(u + x, v + y)$
3. **New measure** for the **presence** of a corner



HARRIS DETECTOR

Mathematical formulation:

- We are looking for a feature that for a slight shift in any direction corresponds to a large difference
- We aim the following energy function to be large for **any** shift $\Delta u = [\Delta x, \Delta y]^T$

$$E_{AC}(\Delta u) = \sum_i \underbrace{w(x_i)}_{\text{Window function}} \underbrace{[I_0(x_i + \Delta u) - I_0(x_i)]}_{\text{Shifted intensity}}^2 \underbrace{I_0(x_i)}_{\text{Intensity}}$$

Sum over a window centered at the point being examined, x_i

$$E_{m,n}(x, y) = \sum_{u,v} w_{m,n}(u, v) [\underbrace{f(u + x, v + y)}_{\text{shifted intensity}} - f(u, v)]^2 \quad \forall m, n, x, y$$

HARRIS DETECTOR

The **shifted patch** can be approximated by the linear terms of a Taylor expansion:

$$f(u+x, v+y) \approx f(u, v) + xf_x(u, v) + yf_y(u, v)$$

Then,

$$E_{m,n}(x, y) = \sum_{u,v} \mathbf{w}_{m,n}(u, v) [f(u+x, v+y) - f(u, v)]^2 \approx \sum_{u,v} \mathbf{w}_{m,n}(u, v) [xf_x(u, v) + yf_y(u, v)]^2$$

where $f_x(u, v) = \partial f / \partial x$ and $f_y(u, v) = \partial f / \partial y$

Taylor Series:

$f(x)$ can be represented at point a in terms of its derivatives

$$f(x) = f(a) + (x-a)f_x + \frac{(x-a)^2}{2!} f_{xx} + \dots$$

HARRIS DETECTOR

$$E_{m,n}(x, y) \approx \sum_{u,v} w_{m,n}(u, v) [x f_x(u, v) + y f_y(u, v)]^2$$

The equation can be written in matrix form as: $E_{m,n}(x, y) \approx [x \ y] M \begin{bmatrix} x \\ y \end{bmatrix}$ for the (m, n) pixel and (x, y) shift

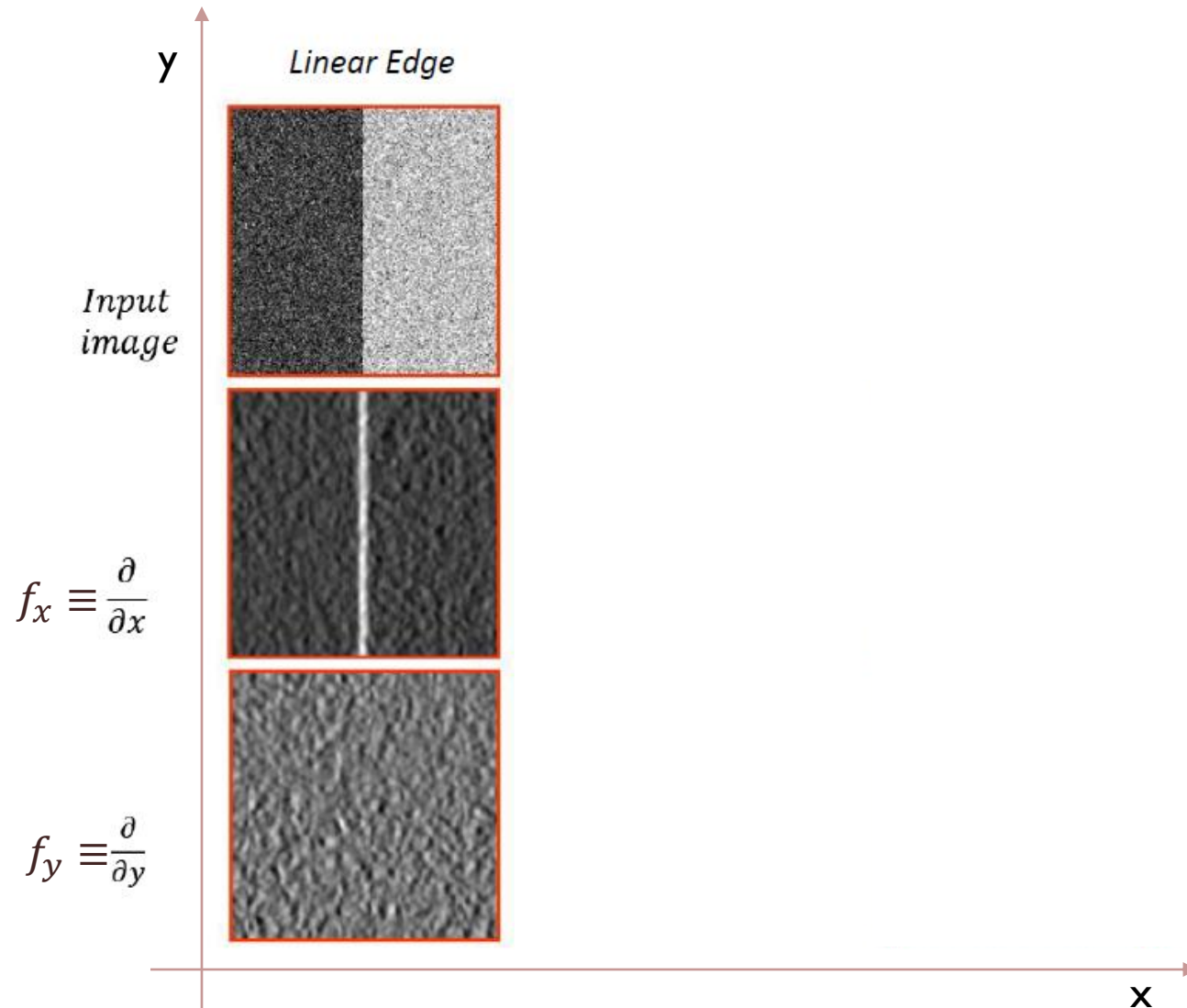
where $M = \sum_{u,v} w_{m,n}(u, v) \begin{bmatrix} f_x^2 & f_x f_y \\ f_x f_y & f_y^2 \end{bmatrix}$



Window function Products of components of the **gradient**

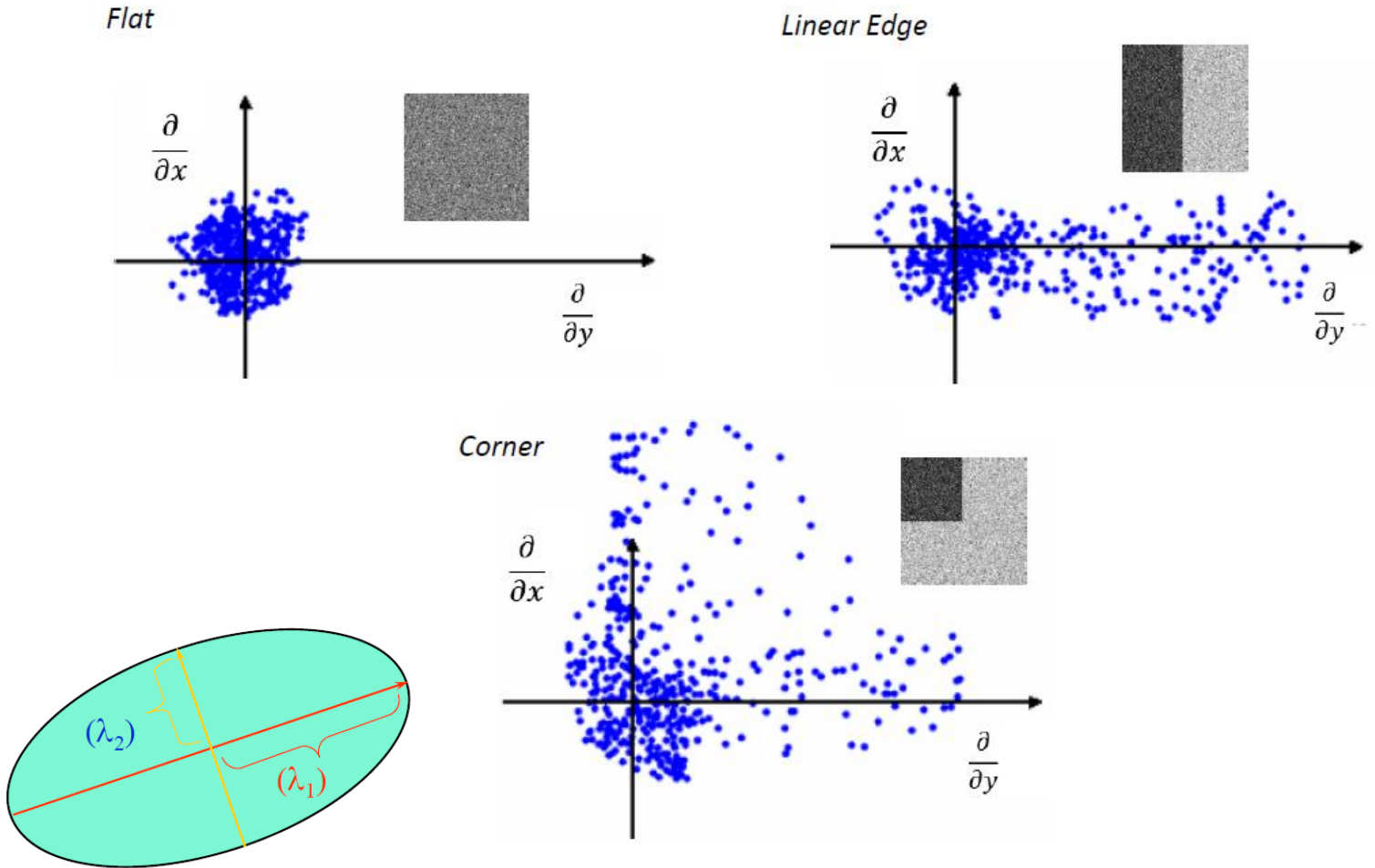
Matrix M is called **Harris matrix**

HARRIS DETECTOR



HARRIS DETECTOR

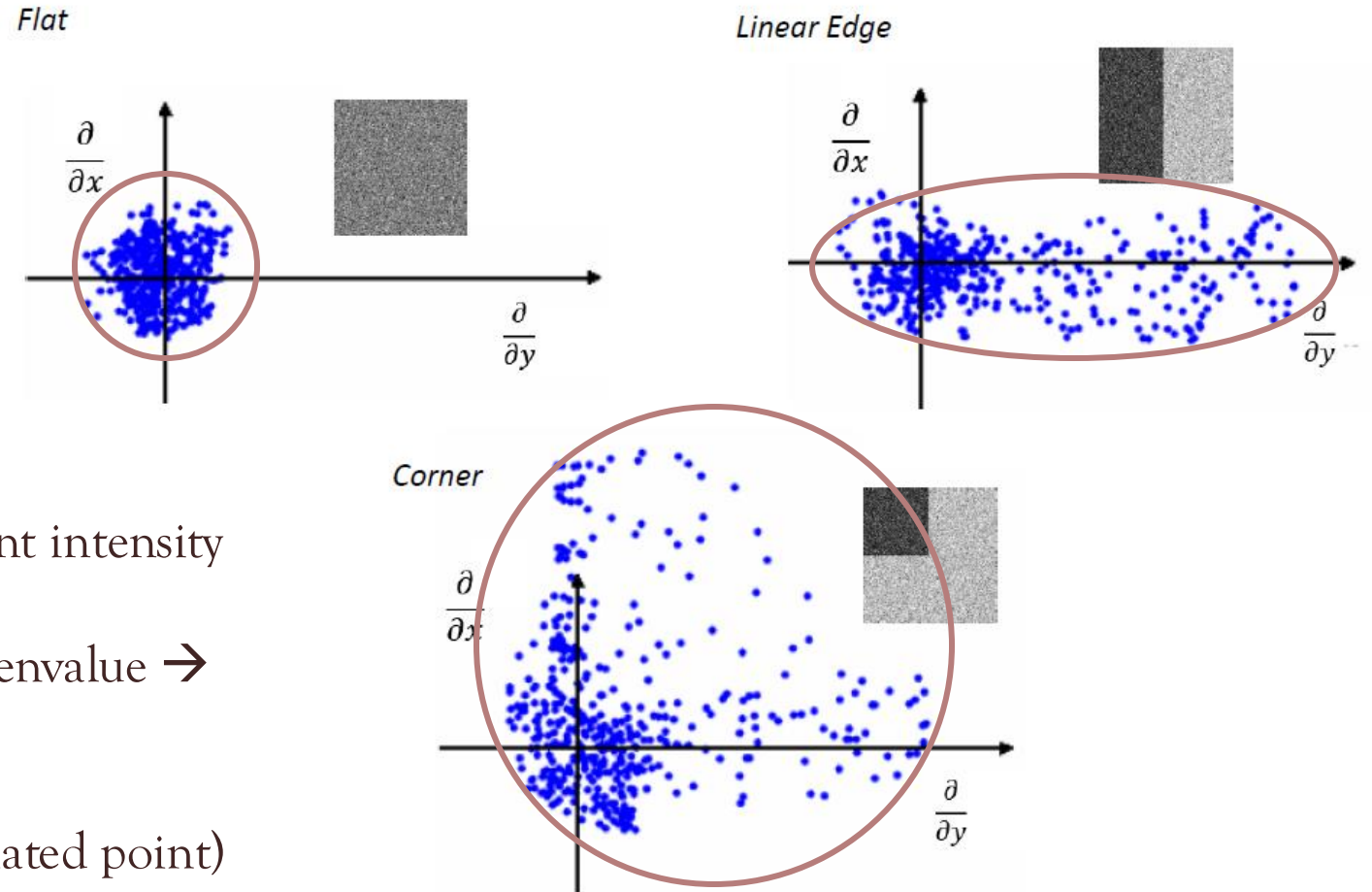
- The **eigenvectors** of a real, symmetric matrix (such as **M**) point in the direction of maximum data spread
- The corresponding **eigenvalues** are proportional to the amount of data spread in the direction of the eigenvectors
- The eigenvectors are the **major axes of an ellipse fitting the data**
- The magnitude of the eigenvalues (λ_1 , λ_2) are the distances from the centre of the ellipse to the points where the ellipse intersects the major axes



HARRIS DETECTOR

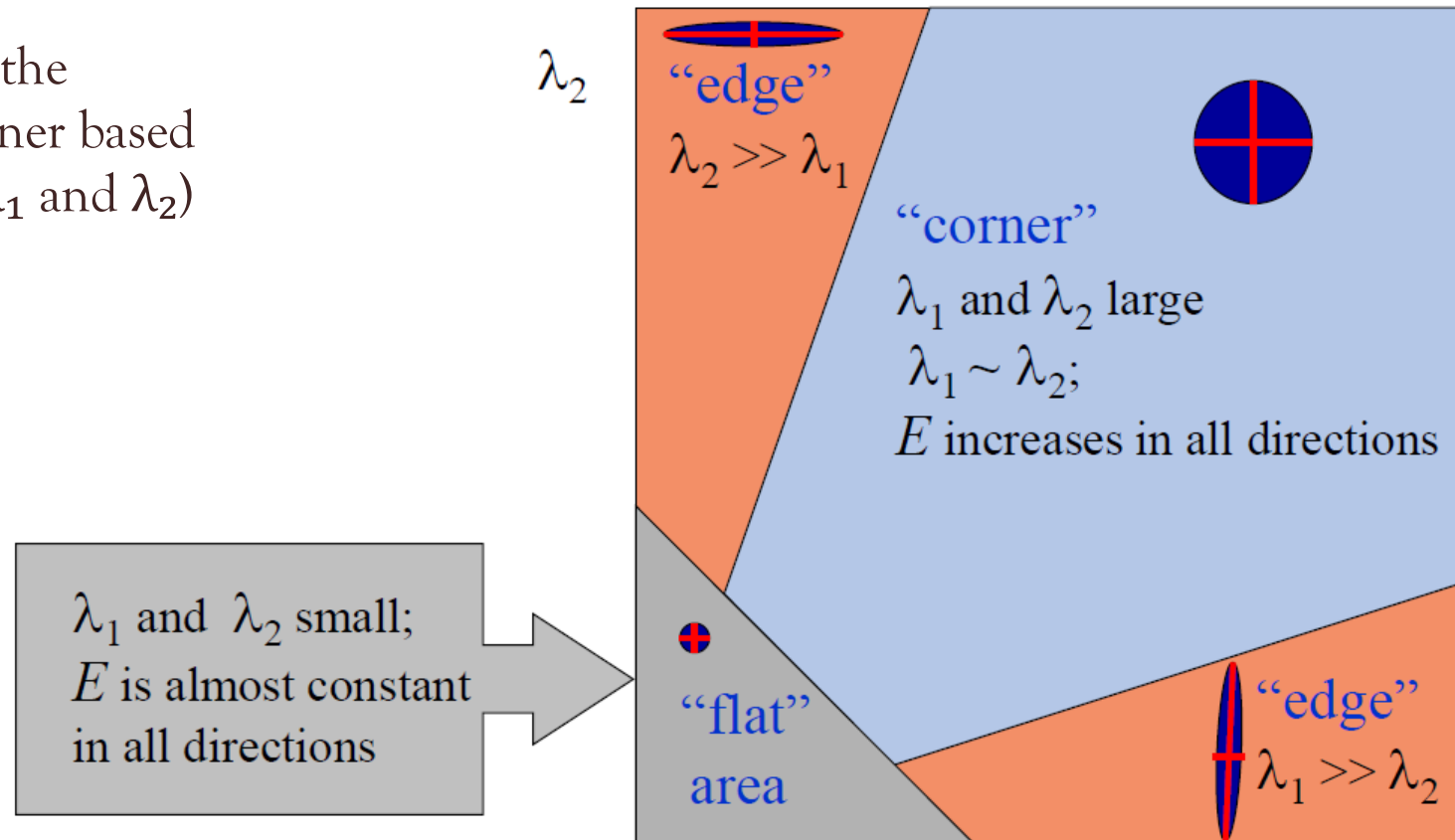
Eigenvectors and eigenvalues (λ_1, λ_2) of the matrix $\rightarrow$ ellipses surrounding the points

1. Two small eigenvalues $\rightarrow$ nearly constant intensity
2. One small eigenvalue and one large eigenvalue $\rightarrow$ vertical or horizontal boundary
3. Two large eigenvalues $\rightarrow$ corner (or isolated point)



HARRIS DETECTOR

New **measure** of the presence of a corner based on eigenvalues (λ_1 and λ_2) of the matrix



HARRIS DETECTOR

- Computation of the **eigenvalues** can be difficult and expensive (squares & square roots)
- Harris detector uses a measure of corner response, **R** , based on the fact that, if matrix M is square:
 $trace(M) = \lambda_1 + \lambda_2$, and $\det(M) = \lambda_1 \lambda_2$

- Then,

$$R = \lambda_1 \lambda_2 - k(\lambda_1 + \lambda_2)^2 = \det(M) - k (trace(M))^2$$

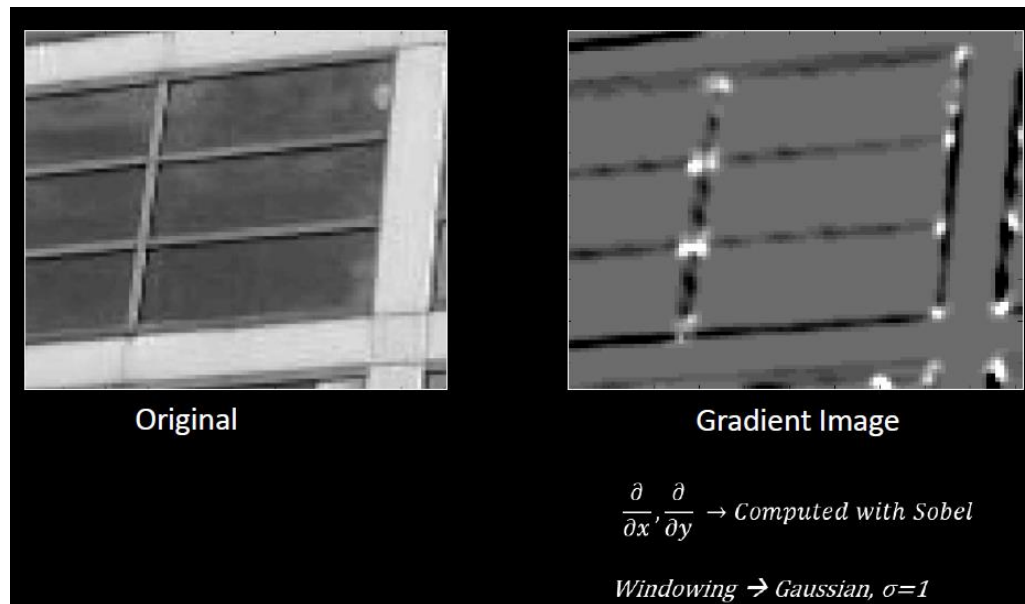
where, **k** is an empirically determined constant

- Optimal **k** value depends on the implementation, typically $0 < k < 0.25$ (some authors suggest $0.04 < k < 0.06$)
- It can be interpreted as a **sensitivity factor**: the smaller it is, the more likely the detector will find corners

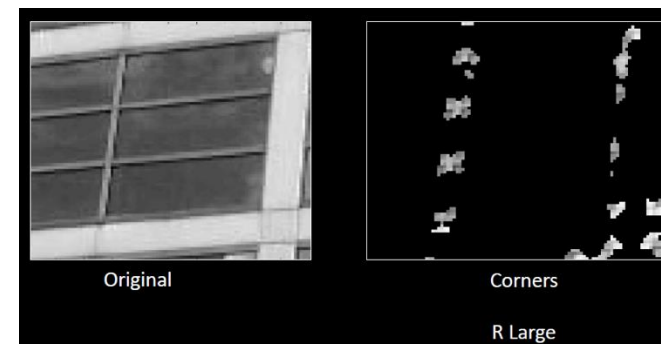
HARRIS DETECTOR

- Measure R has:
 - **large positive** values when both eigenvalues are large $\rightarrow$ **corner**
 - **large negative** values when one eigenvalue is large and the other small $\rightarrow$ **edge**
 - **small** absolute value when both eigenvalues are small $\rightarrow$ **flat region**
- R is typically used with a threshold, T
- **Corner** at an image location has been detected if $R > T$ for a patch at that location

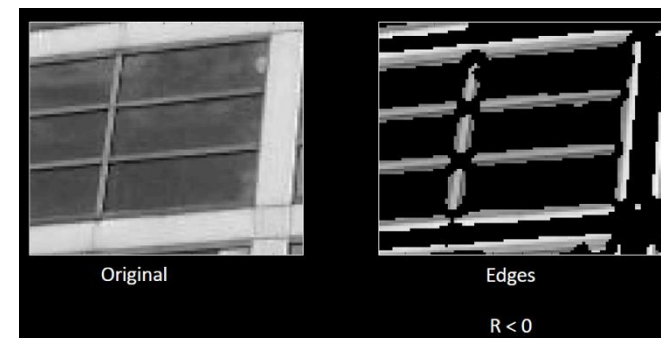
HARRIS DETECTOR: EXAMPLE



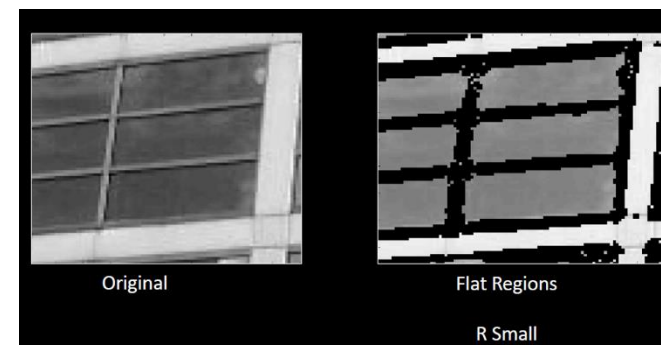
Large positive R
 $\rightarrow$ corners



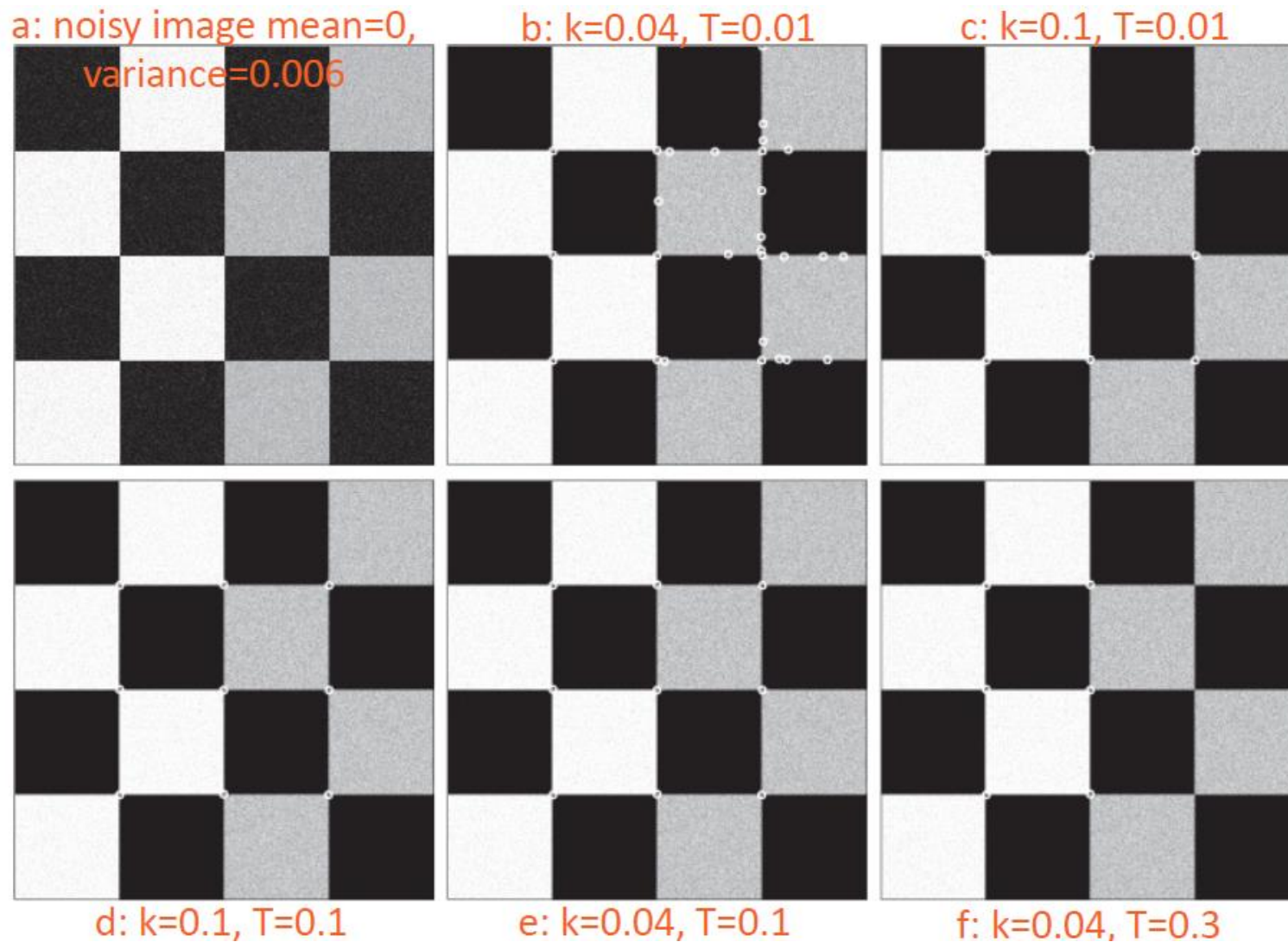
Large negative R
 $\rightarrow$ edges



Small R $\rightarrow$ flat



HARRIS DETECTOR: EXAMPLE



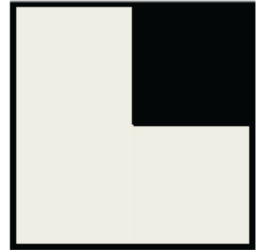
BLOB DETECTOR: MSERS

- Harris corner detector useful in applications characterized by **sharp transitions** of intensities
- The **Maximally Stable Extremal Regions**, MSERs (Matas et al., 2002) are more “blob” **oriented** and are intended to yield whole image features to establish correspondence between two or more images.

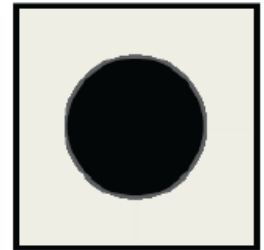
225	175	90	90
125	5	90	225
5	5	5	225
125	100	225	225

- A grayscale image can be viewed as a topographic map, with the xy-axes representing spatial coordinates, and the z-axis representing intensities.
- If we start **thresholding** a (8-bit) greyscale image **one intensity level at a time**, the result of each thresholding is a binary image in which we show the pixels $\geq$ the threshold in white, and the pixels below the threshold as black. If $T=0$, the result is a white image (all pixel values are ≥ 0)
- As we start increasing T , we will begin to see black components in the output binary images. These correspond to local minima in the topographic map view of the image. **Black regions may begin to grow and merge**, but they never get smaller from image to image
- At $T=255$, the resulting image will be black
- Because each stage of thresholding results in a binary image, there will be one or more connected components in each image (in WHITE). The set of all such components resulting from all thresholdings is the set of extremal regions
- Extremal regions that **do not change size** (number of pixels) **over a range of threshold values** are called **maximally stable extremal regions**

Corners



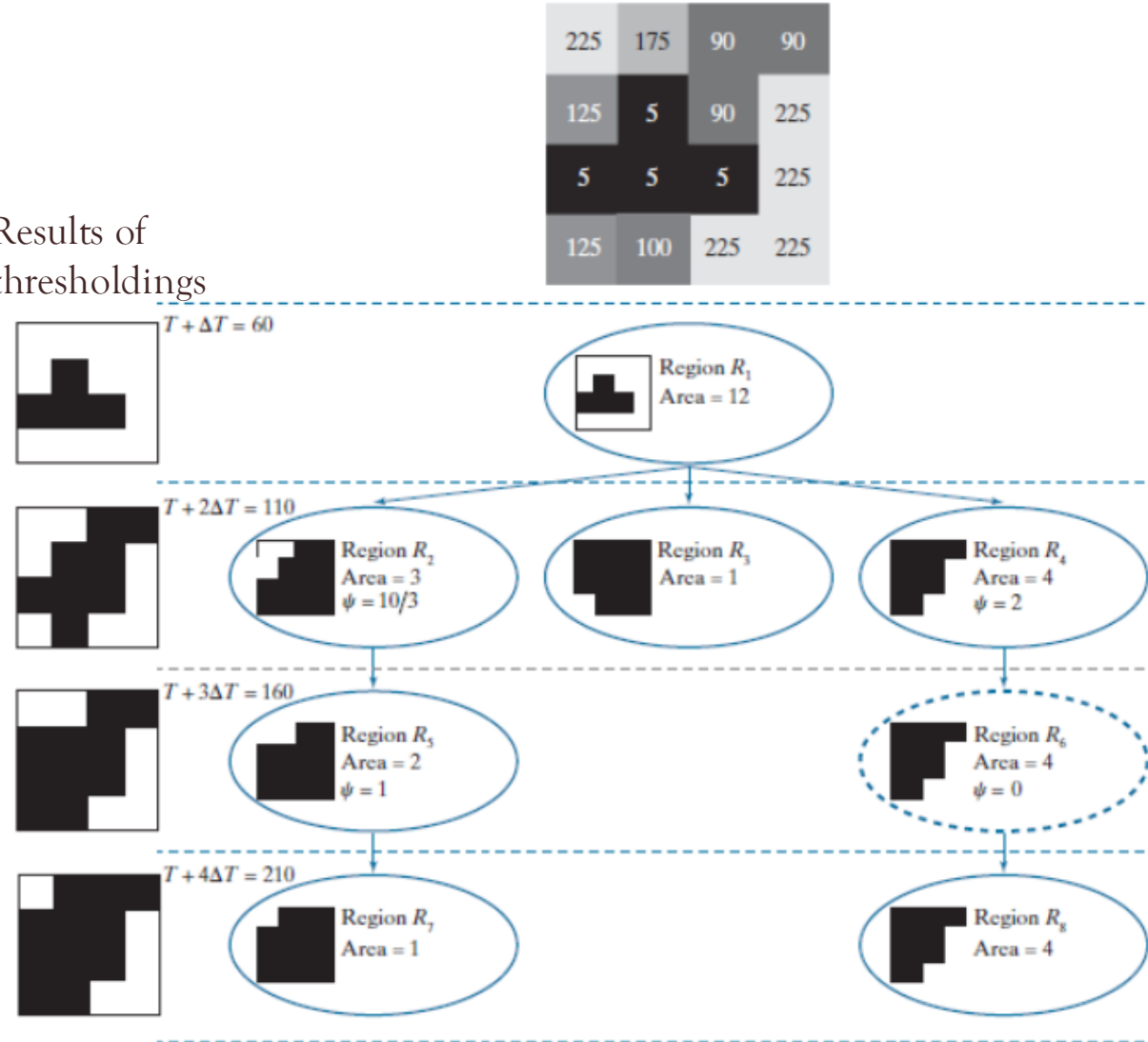
Blobs



BLOB DETECTOR: MSERS

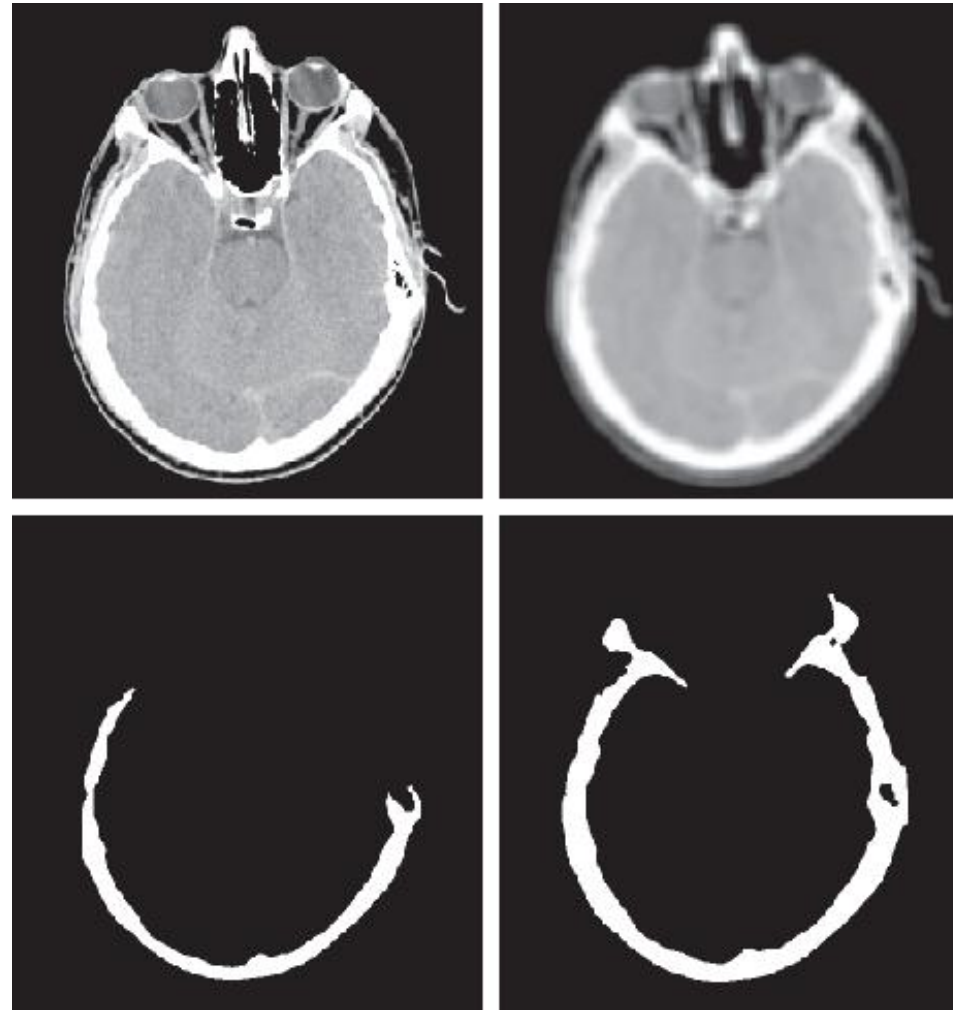
1. Several thresholdings
 2. Check connected components (white)
 3. Build **component tree**
- Check **size variations** between parent and child (contained) regions to determine stability
 - Only region 6 has a parent and child of similar size (identical in this case)
 - **Region 6 is the only MSER detected in this case**

Results of thresholdings



BLOB DETECTOR: MSERS

Detecting MSER from other grayscale images:

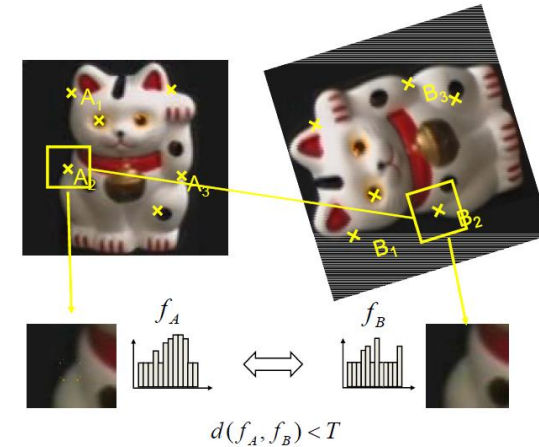


MATCHING ALGORITHMS

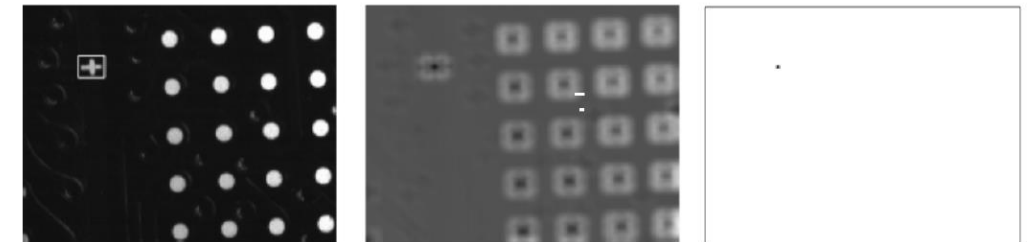
- SIFT for matching features
- This type of process can be used in applications such as:
 - **finding correspondences** between two images for the purpose of image registration
 - finding instances of an image in a database of image

Template matching:

- Find objects in an image by showing the system a prototype of the class of objects
- The object to be found is described by a **template** image
- The **similarity** between the template and parts of the image is computed. Several similarity metrics:
 - Sum of absolute differences (SAD)
 - Sum of squared differences
 - Normalized cross-correlation (invariant to linear illumination changes)
 - Etc.
- If the similarity is high (small difference), an instance of the template is found



K. Grauman, B. Leibe



Location of a fiducial mark (indicated by the white rectangle and used as template) in a PCB using template matching and SAD measure.

Source: Steger C., Ulrich M. Wiedemann C; *Machine Vision Algorithms and Applications*

MATCHING ALGORITHMS: SIFT

- Python code → Lab tomorrow

